Carolina Pacheco Oñate

PhD in Biomedical Engineering
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Biomedical engineer excited about the power of interdisciplinary work for the development of effective solutions to real-world medical needs. With an electrical engineering background, my research interest lies at the intersection of machine learning, computer vision, and biomedical sciences, which I have explored through projects ranging from the cellular scale to full-body motion analysis. I believe that technical expertise along with collaboration with domain experts are fundamental to make meaningful progress in complex biomedical tasks.

Academic Background

2018-2024 **PhD in Biomedical Engineering**, *Johns Hopkins University*, USA

2016-2018 **MSE in Biomedical Engineering**, *Johns Hopkins University*, USA

2008-2014 **Diploma in Electrical Engineering**, *University of Chile*, Chile

2008-2014 **Bachelor of Engineering Science in Electricity**, *University of Chile*, Chile

Professional Background

2017-2024 **Research Assistant**, *Johns Hopkins University (7 years)*

Led and supported several computer vision research projects focused on addressing medical needs. Collaborated in interdisciplinary teams, supported grant writing, and also mentored high school and undergraduate students.

2016 **Project Engineer**, *TrainFES (4 months)*

Designed and deployed an automatic control system for gait support of *foot drop* patients by means of functional electrical stimulation devices.

2014-2015 **Research Engineer**, *CIGIDEN*, (1 year and 5 months)

National Research Center for Integrated Natural Disaster Management
Developed models of pedestrian evacuation and provision of transient accommodations to study the response capacity of public and private sectors in the event of a catastrophe.

2013 **Research Assistant**, *PUC (2 months)*

Intern at "An Intelligent Tutoring System Based on Novel Sensing Modalities" project. Supported the acquisition of electroencephalographic (EEG) recordings and conducted a feasibility study of attentional level detection through EEG signals analysis.

2011 **Research Assistant**, *University of Chile (1 year)*

Biomedical Laboratory, Department of Electrical Engineering
Research assistant at "Sueño" project, which consisted of analyzing brain electrical activity during sleep. The research focus was on the detection of Cyclic Alternating Pattern (CAP).

Selected Projects

2020-2024 **Computer vision and lens-free imaging for automatic monitoring of infection**

Development of computer vision and machine learning algorithms to improve image quality of holographic reconstruction, and perform multi-object, large-scale detection of abnormalities such as bacteria, blood cells, and crystals in urine samples from lensless imaging devices. The main insight is that the incorporation of domain knowledge into deep architectures can alleviate the need for detailed annotations.

2018-2024 **Imitation assessment on children with autism**

Design of a dynamic time warping-based approach to quantitatively assess the imitation performance of children in a videogame-like task. The proposed method shows better predictive capacity of autism diagnosis than human-based imitation assessments.

2018-2019 **GEARing smart environments for pediatric rehabilitation**

Development of activity recognition methods to enable child-robot interaction in a pediatric rehabilitation setting. Design of a detection-based multiview action classification scheme that relies on local features of spatial regions of interest. Leverage of detection scores by novel learnable fusion coefficients to weigh the importance of each view in the final prediction.

2017-2018 **Classification of stem cell-derived cardiomyocytes**

Development of machine learning algorithms to classify action potentials of stem cell-derived cardiomyocytes for which annotations do not exist. Use of recurrent neural networks and concepts such as semi-supervised learning and domain adaptation led to significant advantages with respect to state-of-the-art methods in terms of computational efficiency and accuracy.

Publications

2024 **C. Pacheco**, F. Yellin, R. Vidal and B. D. Haeffele *Vertex Proportion Loss for Multi-Class Cell Detection from Label Proportions*, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Marrakesh, Morocco, 2024.

2023 G. N. McKay, A. Oommen, **C. Pacheco**, M. T. Chen, S. C. Ray, R. Vidal, B. D. Haeffele and N. J. Durr *Lens Free Holographic Imaging for Urinary Tract Infection Screening*, Transactions in Biomedical Engineering.

2022 **C. Pacheco**, G. N. McKay, A. Oommen, N. J. Durr, R. Vidal and B. D. Haeffele *Adaptive sparse reconstruction for lensless digital holography via PSF estimation and phase retrieval*, Optics Express.

2022 G. N. McKay, A. Oommen, **C. Pacheco**, M. T. Chen, S. C. Ray, R. Vidal, B. D. Haeffele, and N. J. Durr *Towards real-time urinalysis with holographic lens-free imaging*, Advanced Biomedical and Clinical Diagnostic and Surgical Guidance Systems.

2021 D. E. Lidstone, R. Rochowiak, **C. Pacheco**, B. Tuncgenc, R. Vidal and S. H. Mostofsky *Automated and scalable Computerized Assessment of Motor Imitation (CAMI) in children with Autism Spectrum Disorder using a single 2D camera: A pilot study*, Research in Autism Spectrum Disorders.

2020 B. Tuncgenc, **C. Pacheco**, R. Rochowiak, R. Nicholas, S. Rengarajan, E. Zou, B. Messenger, R. Vidal and S. H. Mostofsky *Computerised Assessment of Motor Imitation (CAMI) as a scalable method for distinguishing children with autism*, Biological Psychiatry: Cognitive Neuroscience and Neuroimaging.

2020 **C. Pacheco**, E. Mavroudi, E. Kokkoni, H. G. Tanner and R. Vidal, *A detection-based approach to multiview action classification in infants*, International Conference on Pattern Recognition (ICPR), 2020.

2019 **C. Pacheco** and R. Vidal, *An unsupervised domain adaptation approach to classification of stem cell-derived cardiomyocytes*, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Shenzhen, China, 2019.

2018 **C. Pacheco** and R. Vidal, *Recurrent neural networks for classifying human embryonic stem cell-derived cardiomyocytes*, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), Granada, Spain, 2018.

2017 **C. Pacheco**, M.A. Duarte-Mermoud, N. Aguila-Camacho and R. Castro-Linares, *Fractional-order state observers for integer-order linear systems*, Journal of Applied Nonlinear Dynamics, 6 (2): 251-264, 2017.

2015 **C. Pacheco**, P. Karelovic and A. Cipriano, *Comparing dynamic evacuation control strategies for emergency situations*, European Control Conference, Linz, Austria, 2015.

Teaching and mentoring

2019-2022 Research practicum mentor of high-school students from the Ingenuity Project, a program for talented, high-achieving, Baltimore City public school students.

2018-2019 Volunteer for the STEM achievement in Baltimore Elementary Schools (SABES) program, which seeks to improve STEM curriculum and delivery in grades 3 5.

2009-2018 Teaching assistant of Biomedical Data Science (Johns Hopkins University, Fall 2018) and several other courses at University of Chile (Advanced Control Laboratory, Optimization in Control Systems, Linear Algebra, Single Variable Calculus, among others).

Honors and Awards

2024 **MICCAI NIH Participation Award**

This award provides financial support to US-based graduate students to assist the conference. The student must be first author of an accepted paper.

2022-2023 **Amazon AI2AI Fellowship**

This award provides year-long funding to PhD students from Johns Hopkins University, based on their outstanding publication record, research proposal, and mentor support.

2019 **MICCAI Graduate Student Travel Award**

This award provides financial support to world-wide graduate students to assist the conference. The student must be first author of an accepted paper.

2016-2018 **Becas-Chile Scholarship for Master studies**

This program supports competent Chilean students to pursue graduate studies abroad. Its benefits include round trip tickets, monthly tuition fees, among others.

2012 **Best student of the year**

The Department of Electrical Engineering of University of Chile rewarded best students of each level based in their academic performance.

Languages

Spanish - Native speaker

English - Full professional proficiency

Competences and Skills

Software/Coding - Python, Pytorch, LateX, Matlab, Excel.
Skills - Interdisciplinary collaboration, research, computer vision, scientific writing.

Last Update: September, 2025.